

Ahmed Zaheer

Professional Summary

Full-stack developer specializing in production AI/ML systems. Collaborate on building end-to-end LLM applications - agentic tool-calling dashboards, multi-turn conversation systems, and AI-driven analytics - as well as generative-image infrastructure including GPU-accelerated worker orchestration, multi-model dataset quality pipelines, and diffusion model editing workflows. Strong across Next.js, NestJS, React Native (Expo), MongoDB, SQL, Redis/BullMQ, and cloud infrastructure (GCP and AWS). Equally comfortable shipping polished user-facing features and the reliability work behind them; schema design, error recovery, observability, and performance optimization. Self-motivated and detail-oriented, with a passion for turning research-grade AI tooling into reliable production systems.

Experience

DesignGurus - 2025 - Present

AI/ML Engineer (cross-functional role with Full-Stack Development)

Melbourne, Victoria, Australia | Remote

- Built a multi-model dataset quality analyzer for image-training pipelines using Xenova Transformers - combining vision-transformer feature extraction (SigLIP), object detection (DETR), Laplacian-variance sharpness scoring, cosine-similarity duplicate detection, and face-similarity clustering - with concurrent batch inference and weighted multi-metric scoring (~1,200 lines).
- Engineered an image-editing dashboard for diffusion model workflows with variant-aware workflow templates, client-side node-graph rewiring for multi-image conditioning, and real-time job status tracking.
- Built a GPU-accelerated NestJS worker orchestration layer for AI job processing - self-healing training-job recovery with exponential retry, workflow sanitization, cloud-storage model upload validation, and bulkWrite optimization that collapsed N database queries into a single batched update.
- Partnered in the design and deployment of GPU-accelerated containerized ML infrastructure (NVIDIA CUDA + PyTorch) with a two-stage Docker build strategy separating 100GB+ model and dependency layers from application code; hardened cross-platform image upload pipelines (Next.js web + Expo/React Native mobile) with buffer validation, ReDoS-safe CORS middleware, and memory-leak fixes.

DesignGurus - 2024

Full-Stack Developer

Melbourne, Victoria, Australia | Remote

- Developed and shipped a multi-tool AI command center for admins within an agile environment, utilizing Google Gemini function calling - 30+ tools spanning sales analytics, user metrics, content quality, and fraud/security detection; implemented conversational state,

multi-turn history management, saved filters, and shareable read-only links (~5,000 lines across Next.js frontend and Node.js/Express backend).

- Built sophisticated MongoDB aggregation pipelines powering the AI tools - multi-stage \$facet rollups for revenue and bundle allocation, subscription cohort analysis, and security forensics (impossible-travel detection, suspicious-account discovery, IP-sharing analysis).
- Built an AI-powered Roadmapper using Gemini with document tagging, Borda-score filtering, and role-to-difficulty mapping; implemented 30-day TTL caching and daily generation limits to control LLM costs.
- Led LLM model migrations across the codebase - OpenAI gpt-4 to gpt-5-nano and Gemini 3.1 Flash Lite upgrades - with regression testing, prompt updates, and downstream tool-call compatibility fixes.
- Shipped a flashcard authoring system: schema design, bulk upsert with race-condition (E11000) handling, JSON import flow, and flip-card player rendering in the course playback experience.
- Designed admin tooling including a Next.js ISR cache revalidation dashboard with multi-URL parsing and per-path success/failure reporting, plus SEO-gated content export APIs for articles and courses.
- Diagnosed and fixed reliability issues in AI chat infrastructure - history payload bloat across stored conversations, tool-call history loss on errors, and graceful handling of agent iteration limits and Firebase token edge cases.

Key Skills

- **AI & LLM Engineering:** Google Gemini (function calling, agentic tools, multi-turn conversation), OpenAI API (GPT-4 through GPT-5 families, model migrations), Anthropic Claude, prompt engineering, structured outputs, streaming responses, Xenova Transformers (SigLIP, DETR), diffusion model workflows, LoRA fine-tuning, Hugging Face Hub
- **Cloud & Infrastructure:** AWS, Google Cloud Platform (GCS), Docker (multi-stage GPU-enabled builds), Docker Compose, NVIDIA Container Toolkit (CUDA, PyTorch), Kubernetes, GitHub Actions CI/CD, Cloud Security
- **Frontend & Mobile:** Next.js 14/15 (App Router + Pages), React 19, TypeScript, Material-UI, Tailwind CSS, SWR, ApexCharts, React Native, Expo, Expo Router, NativeWind, React Query
- **Backend & Languages:** Node.js, Express, NestJS, Python, TypeScript, JavaScript, HTML, CSS, MongoDB (Mongoose, aggregation pipelines, bulkWrite), Redis, BullMQ, Kafka, MySQL, PostgreSQL, Firebase Auth, REST API design
- **Tooling & General:** Git, Linux, Cypress E2E testing, PostHog, Sentry, Microsoft Office, Application UI design, role-based access control, schema design and migrations

Education

- **VCE Graduate**
Mount Ridley College
- **Diploma of IT**
Victoria University